STANDALONE DATA TABLES AND FIGURE LEDGER: PAPER I (v2.1 - FED)

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PART I: STANDALONE DATA TABLES

Table 1: Physical Parameters, Material Baselines, and Operational Safety Factors across Fast-Neutron Fluence (0-20 dpa).

[Verified SI Units: MW/m^2, kPa, dpa, MPa, K, A/mm^2, T, mol/m^2.s]

Operational State	Fluence (dpa)	Pore Radius r_p (um)	Capillary Press. Delta P_cap (kPa)	Permeability K (10^-11 m^2)	Dryout Flux q''_dryout (MW/m^2)	Safety Factor (SF)	Operational Status
Pristine State	0.0	25.00	27.05	1.25	41.50	2.77	Baseline Optimal
Replacement Limit	9.8	17.54	38.55	0.60	22.50	1.50	Mandatory Replacement
Dryout Parity	13.0	16.49	41.00	0.50	15.00	1.00	Thermal Dryout Limit
Extreme Damage	20.0	15.20	44.48	0.38	8.69	0.58	Sub-critical Failure

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PART II: STANDALONE FIGURE LEDGER & CAPTIONS

Figure 1: Porous Tungsten CPS Matrix Degradation and Capillary Pressure Rise under Fast-Neutron Radiation (0-20 dpa).

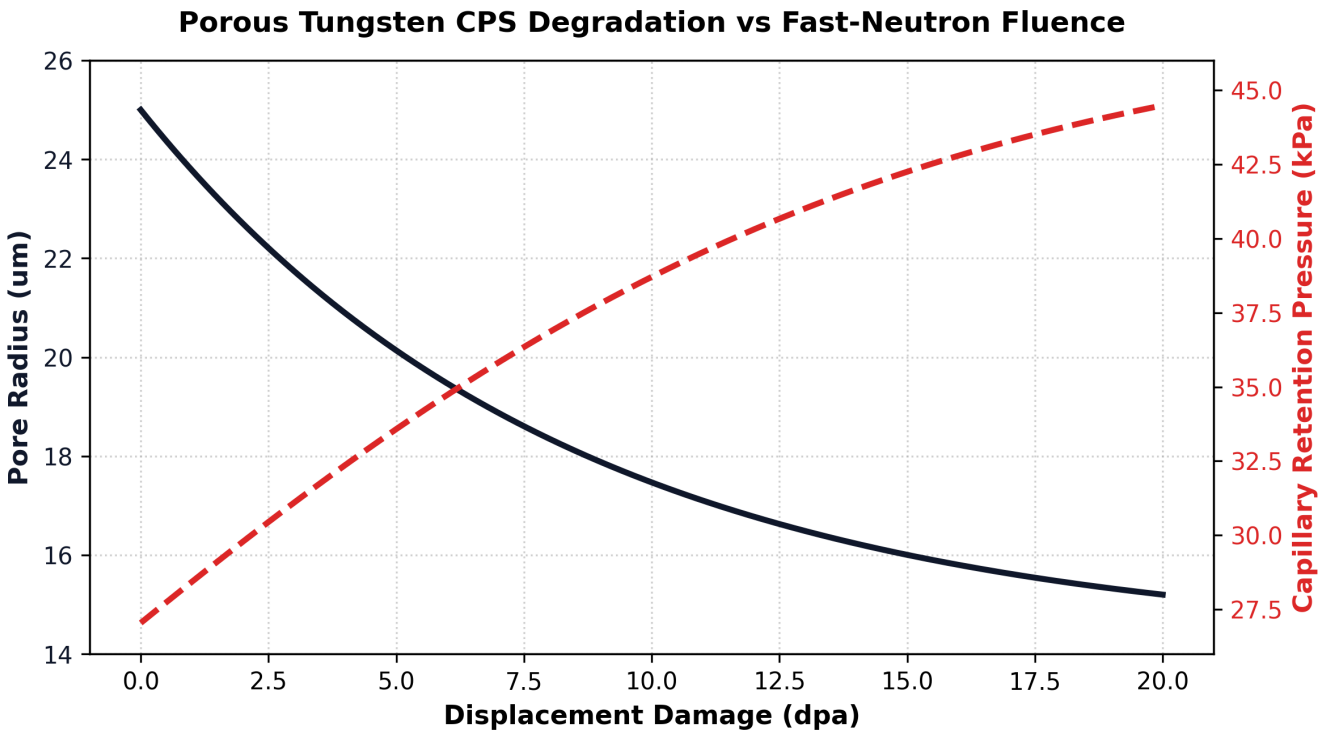


Figure 1: Porous Tungsten CPS Matrix Degradation and Capillary Pressure Rise under Fast-Neutron Fluence (0–20 dpa).

Caption: Fast-neutron displacement damage (0-20 dpa) contracting tungsten pore radius from 25.0 um to 15.2 um (solid navy curve, left axis) while static Young-Laplace capillary retention pressure increases from 27.05 kPa to 44.48 kPa (dashed red curve, right axis).

Figure 2: Non-linear Carman-Kozeny Permeability Collapse and Critical Dryout Heat Flux Degradation with Operational Safety Factor Limits.

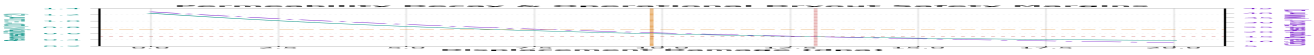


Figure 2: Non-linear Carman-Kozeny Permeability Collapse and Critical Dryout Heat Flux Safety Thresholds (9.8 dpa vs 13.0 dpa).

Caption: Fast-neutron fluence driving a 70% collapse in matrix permeability K (dashed navy curve, left axis) and reducing critical dryout heat flux q''_dryout from 41.50 MW/m^2 down to sub-critical levels (solid red curve, right axis). Mandatory engineering replacement safety factor threshold (SF = 1.50, q''_dryout = 22.50 MW/m^2) is crossed at 9.8 dpa, while complete thermal dryout parity with 15.0 MW/m^2 baseline divertor heat load (SF = 1.00) occurs at 13.0 dpa.